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Shock tube ignition delay time measurements in propane/O_2/argon mixtures at near-constant-volume conditions

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Abstract

Shock tube measurements of ignition delay times with high activation energies are strongly sensitive to variations in reflected shock temperatures. At longer shock tube test times, as are needed at low reaction temperatures, small gradual increases in pressure (and simultaneous increases in temperature) that result from incident shock wave attenuation and boundary layer growth can significantly shorten measured ignition delay times. To obviate this pressure increase, we made use of a recently developed driver-insert method of Hong et al. [8] that allows generation of near-constant-volume test conditions for reflected shock measurements. Using this method, we have measured propane ignition delay times in a lean mixture (0.8% C_3H_8/8% O_2/Ar) over temperatures between 980 and 1400 K and nominal pressures of 6, 24 and 60 atm, under both conventional shock tube operation (with post-shock fractional pressure variation $dP_5/dt \sim 1$–7%/ms) and near-constant-volume operation (with $dP_5/dt \sim 0$%/ms). The near-constant-volume ignition delay times provide a database for low-temperature propane model development that is independent of non-ideal fluid flow and heat transfer effects. Comparisons of these near-constant-volume measurements with predictions using the JetSurF v1.0 mechanism of Sirjean et al. [10] and the Curran et al. mechanism of NUI Galway [5] were performed. Ignition delay times measured with $dP_5/dt \sim 1$–7%/ms were found to be significantly shorter (about 1/3 of the near-constant-volume values) at the lowest temperatures and highest pressures studied. However, these ignition times are successfully simulated using the JetSurF v1.0 mechanism when an appropriate gasdynamic model that accounts for changes in pressure and temperature is used.
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Keywords: Shock tube; Ignition delay time; Propane–oxygen; Constant volume

1. Introduction

Recently, there has been an increased interest in propane ignition data at high pressures and low temperatures due to their significance in the validation of detailed chemical kinetic mechanisms at these conditions. Both shock tubes and rapid compression machines (RCM) have been used for these studies. Cadman et al. [1] examined auto-ignition delay times in shock-heated lean propane–air mixtures ($\Phi = 0.5$) in the low-temperature region of 835–1400 K at pressures of 5–40 bars. Similar shock tube studies, performed by Herzler et al. [2] and Petersen et al. [3] at temperatures of

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750–1300 K and pressures of 10 and 30 bars, are in agreement with the measurements by Cadman et al. These shock tube ignition delay times start to deviate (roll-off) from detailed model simulations at temperatures around 1000 K. More recently, Gallagher et al. [4] studied propane oxidation in air using an RCM over temperatures between 680 and 970 K at different compressed gas pressures and equivalence ratios. The measured RCM ignition delay times are approximately two orders of magnitude longer than the shock tube data, and a characteristic negative coefficient behavior was observed. Of critical importance is the fact that modeling of the ignition delay times using a detailed propane oxidation mechanism (Curran et al. [5]) under commonly employed constant internal energy (U) and volume (V) constraints cannot reproduce the measurements at lower temperatures for either shock tube or RCM data (see Fig. 1).

A similar discrepancy has been observed in recent reflected shock studies of hydrogen–oxygen–argon mixtures. Pang et al. [6] found that hydrogen ignition delay times are at least an order of magnitude shorter than the constant U, V predictions of detailed kinetic mechanisms at temperatures less than 930 K. For these low-temperature experiments, a gradual pressure increase (dP_5/dt) was observed. That study concluded that the post-shock pressure variation contributed significantly to shorter ignition delay times in long-duration (low-temperature) experiments. In an ideal shock tube experiment, the incident shock travels at a constant speed and is reflected from the driven-section endwall. The reflected shock further compresses the incident-shocked test gas, yielding a stagnant and uniform high-temperature region. Computationally, this reflected shock region is often approximated as a constant volume reactor. Additionally, gases in the reflected shock region are often assumed to have constant internal energy

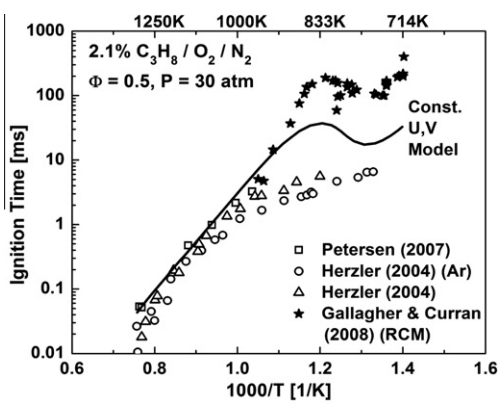

Fig. 1. Previous ignition delay time measurements for propane oxidation in air at $\Phi = 0.5$. The constant U, V model calculations utilize the Curran et al. mechanism [5].

(U), as would be appropriate for adiabatic systems without work addition or chemical energy change. However, due to viscous effects, sidewall boundary layers are developed behind incident shock waves, causing incident shock attenuation and leaving a non-uniform flow field in the incident shock region. As the reflected shock propagates into this flow field and interacts with the growing boundary layers, changes in the post-reflected shock pressure may occur that can penetrate into the reflected shock region [7]. Consequently, an increase in pressure (and a concomitant increase in temperature) is typically observed in the reflected shock region, introducing uncertainties in the ignition delay time measurements. In such experiments, the constant-volume assumption is not strictly valid.

Furthermore, the undiluted mixtures at elevated pressures exhibit an accelerated ignition caused by some localized pre-ignition energy release, resulting in a dramatic and sudden pressure rise prior to the primary ignition event (as discussed by Petersen et al. [3]). Such dramatic pressure effect is different from the normal, gradual pressure increase (dP_5/dt) observed in dilute mixtures. Both pressure effects cannot be simply reproduced by a detailed kinetic mechanism under constant U, V constraints. In the current study, we investigate only the effects of the gradual pressure increase on ignition delay time measurements using a dilute mixture. Hence, we measured propane ignition delay times in a lean mixture (0.8% C_3H_8/8% O_2/Ar) at nominal pressures of 6, 24 and 60 atm, first using a conventional shock tube, to confirm the discrepancies between experimental data and simulations based on two current chemical mechanisms using constant U and V constraints. We then employed a driver-insert method (developed in our laboratory [8]) to minimize the post-shock pressure variations and generate near-constant-volume test conditions for the propane oxidation study. (Preliminary 6-atm propane ignition time measurements without the driver insert were previously described in Ref. [9].) These near-constant-volume measurements were compared to constant U, V predictions using the JetSurF v1.0 mechanism of Sirjean et al. [10] and the Curran et al. mechanism of NUI Galway [5]. Finally, a thermodynamic–gasdynamic model, CHEMSHOCK [11], was used to simulate the conventional shock tube ignition delay data (i.e., with $dP_5/dt > 0$) by incorporating the effects of the measured non-ideal growth of pressure (caused by incident shock wave attenuation and boundary layer growth).

2. Experimental set-up

2.1. Low-pressure shock tube experiments

Ignition delay times at pressures near 6 atm were measured in the Stanford stainless-steel,

high-purity low-pressure shock tube (LPST) previously described by Oehlschlaeger et al. [12]. The shock tube inner diameter is 14.13 cm, and the driven-section is 8.54 m in length, separated from the driver section by polycarbonate diaphragms (0.5 and 1.0 mm in thickness). Under normal shock tube conditions, the driver section is 3.35-m long, and helium is used as the driver gas to obtain a 2-ms high-quality test time. To extend the test time for low-temperature ignition measurements, the driver section was lengthened to 7.12 m to allow more time before the arrival of the rarefaction fan at the test location. The stainless-steel driver section extension has an inner diameter of 14.13 cm and a U-shaped structure with a 23-cm radius of curvature on the centerline. In addition, tailored driver gas mixtures of 30–40% nitrogen in helium were used to minimize reflected shock-contact surface interactions [13]. These modifications provide an available test time of approximately 25 ms at a temperature of about 1000 K.

The gases used, propane (Instrument Grade) 99.5%, oxygen (Research Grade) 99.999%, and argon (Research Grade) 99.999%, were supplied by Praxair and used without further purification. A test gas mixture (0.8% C_3H_8, 8% O_2, with balance argon) was prepared manometrically in a 12-l stainless-steel mixing cylinder and mechanically mixed with a magnetically-driven stirrer for at least 2 h prior to the experiments.

All measurements were performed behind reflected shock waves at a test location 2 cm from the driven-section endwall. Incident shock velocities were measured with five piezo-electric pressure transducers (PCB 113A) spaced axially over the last 1.5 m of the tube and extrapolated to the endwall. The signals of the transducers triggered by the incident shock were delivered to four Philips PM6666 counter timers (with resolution of 0.1 μs), which determined the shock-passage time intervals. The incident shock velocity at the endwall can then be estimated by a linear extrapolation of the incident shock velocities determined from the shock-passage time intervals. Typical shock attenuation rates ranged from 0.8% to 1.5% per meter. Reflected shock conditions were calculated using one-dimensional shock relations and assuming vibrational equilibrium and frozen chemistry. Uncertainty in the initial temperature and pressure was approximately ±0.7% and ±1%, respectively, mainly due to the uncertainty in the measured shock velocity (±0.2%). A Kistler piezo-electric pressure transducer (603B1) coated with RTV silicone rubber was utilized to measure pressure-time history in the test section at 2 cm from the endwall. In addition, OH^* emission chemiluminescence was monitored at the test section using a vertical slit, lens, Schott Glass UG5 filter (with >95% transmission at 306 nm), and modified UV-enhanced Thorlabs PDA36A photodiode detector. The temporal resolution of this emission set-up is typically 10 μs or better.

2.2. High-pressure shock tube experiments

All experiments at pressures higher than 20 atm were performed in the Stanford stainless-steel, high-purity, high-pressure shock tube (HPST). A complete description of this shock tube is provided by Petersen and Hanson [14]. The driver section is 3-m long with a 7.5-cm inner diameter, and the driven-section is 5-m long with a 5-cm inner diameter, separated by an aluminum diaphragm (1.27–3.18 mm in thickness) with cross-scribing. Helium was used for the driver gas to provide approximately 2–3 ms of high-quality test time. At lower temperatures, longer test time was achieved by using tailored driver gas mixtures of 40–50% nitrogen in helium [13], and the available test time was approximately 10 ms at a temperature of about 1000 K. All measurements were made behind reflected shock waves, at a test location 1 cm from the driven-section endwall. The incident shock velocities were measured using six piezo-electric pressure transducers (PCB 113A), with five corresponding Philips PM6666 counter timers (with resolution of 0.1 μs), spaced over the last 2 m of the shock tube and extrapolated to the endwall. Shock attenuation rates varied from 1.0% to 3.5% per meter. Endwall shock velocity, incident and reflected shock conditions were determined using the same method as described above. Uncertainty in the initial temperature and pressure is less than ±1%, with the primary contribution being uncertainty in the measured shock velocity. Pressure–time history in the test section was monitored by a Kistler pressure transducer (603B1) coated with RTV. Additionally, emission from OH^* chemiluminescence was measured at the test location using the similar set-up as described above.

2.3. Driver-insert method

Non-ideal gasdynamic effects of shock waves (primarily incident shock wave attenuation and sidewall boundary layer growth) can cause a gradual increase in pressure (and a simultaneous increase in temperature) in the reflected shock region [9,14,15]. Hong et al. [8] have developed a semi-analytical model for designing shock tube driver inserts to obviate non-ideal pressure variations in the reflected shock region. Optimal configuration for the driver insert uses a parabolic shape, which is dependent on the reflected shock temperature, pressure, and the composition of the driver and driven gases. The driver insert is used to reflect part of the rarefaction fan back to the test section, superimposing a pressure decrease on the non-ideal pressure growth due to the boundary layers. If both effects have the

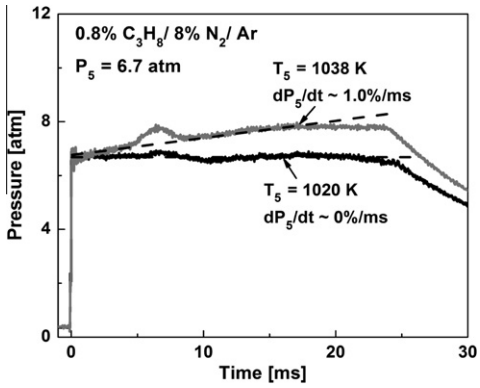

Fig. 2. Comparison of pressure profiles for a mixture of 0.8% C_3H_8/8% N_2/Ar obtained with and without driver insert in the Stanford 14.13 cm diameter shock tube. The fractional pressure rise without driver insert (over 20 ms) is approximately 20%, compared to ±3.0% local pressure variations with driver insert. The decay beginning at 25 ms is due to arrival of the rarefaction wave from the driver section.

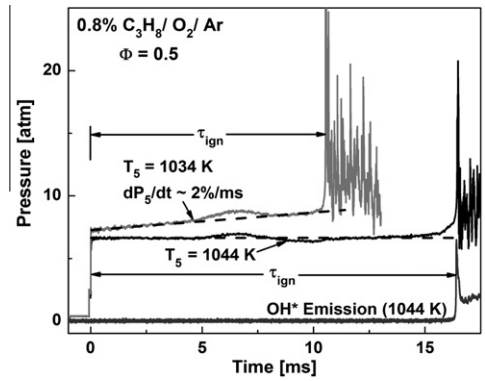

Fig. 3. Comparison of pressure profiles for reactive mixture with and without LPST driver insert. Pressure rise without driver insert (over 10 ms) is 20%, compared to ±3.0% pressure variations with driver insert. Initial reflected shock conditions: $T_5 = 1034$ K and $P_5 = 7.1$ atm (with $dP_5/dt \sim 2\%/ms$), $T_5 = 1044$ K and $P_5 = 6.7$ atm (with $dP_5/dt \sim 0\%/ms$).

same order of magnitude, the non-ideal pressure rise can be effectively eliminated, yielding near-constant-volume test conditions (with a fractional rate of pressure change $dP_5/dt \sim 0\%/ms$) in the reflected shock region (see Fig. 2). With a proper combination of driver insert and tailored driver gas mixtures, long uniform test times of 25 and 10 ms can be acquired at temperatures around 1000 K for the low- and high-pressure shock tubes, respectively. Additionally, constant pressure implies a constant temperature profile by assuming isentropic behavior of the gas at the test section. This has been confirmed with an *in situ* two-line thermometry diagnostic using two distributed feedback (DFB) diode lasers near 2.7 μm [16].

3. Results and discussion

Lean propane ignition delay times were measured at temperatures between 980 and 1400 K and nominal pressures of 6, 24 and 60 atm. The ignition delay time is defined as the time interval between the arrival of the reflected shock and the initial rise in the pressure and excited OH (OH^*) emission traces. Ignition data obtained from the pressure and emission traces are consistent within ±1%. The overall uncertainty in ignition delay time measurements is approximately ±10%, with the primary contribution from the uncertainty in reflected shock temperatures. The typical facility-related rates of pressure change in the LPST and the HPST without the use of driver inserts were 1–3%/ms and 6–7%/ms, respectively. The non-ideal pressure growth is larger in the HPST, primarily due to the fact that its driven-section has a smaller inner diameter. Clearly, the inner diameter of the shock tube can have a significant impact on post-shock pressure variations.

Representative LPST propane ignition data are illustrated in Fig. 3. In the reflected shock wave experiment without the driver insert, a gradual pressure rise with a rate of 2%/ms is observed after the passing of the reflected shock. This non-ideal behavior causes approximately a 20% increase (over 10 ms) in pressure prior to ignition, and concomitantly (assuming isentropic compression) approximately an 8% increase in temperature. Such pressure rise is caused by the incident shock attenuation and boundary layer growth. In the reflected shock wave experiment with a properly designed driver insert (also shown in Fig. 3), the pressure rise behind the reflected shock was eliminated. The pressure profile obtained using the driver insert is nearly flat with local pressure variations of ±3% prior to ignition, relatively insignificant when compared to the 20% pressure rise in the uncompensated conventional experiments. The ignition delay time measured under conventional operating conditions (without the driver insert) at the initial temperature $T_5 = 1034$ K is approximately 7 ms shorter than that obtained under near-constant-volume conditions (with the driver insert) at $T_5 = 1044$ K, owing to the increase in reaction temperature occurring in the former.

Examples of HPST ignition time data with and without driver inserts are shown in Fig. 4. The initial reflected shock pressure of both experiments is approximately the same (~54 atm), and the ignition delay time of the measurement taken without the use of a driver insert ($dP_5/dt \sim 7\%/ms$) at a

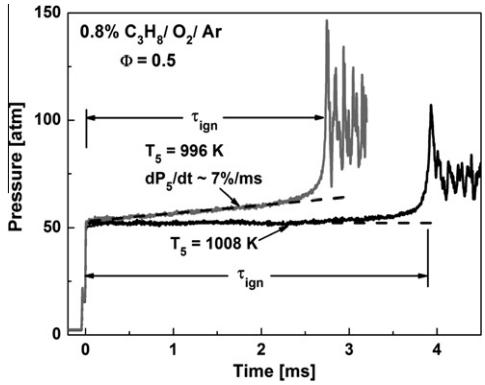

Fig. 4. Comparison of pressure profiles for reactive mixture with and without HPST driver insert. Pressure rise without driver insert (over 2.5 ms) is 17.5%, compared to ±1.0% pressure variations with driver insert. Initial reflected shock conditions: $T_5 = 996$ K and $P_5 = 54.7$ atm (with $dP_5/dt \sim 7\%/ms$), $T_5 = 1008$ K and $P_5 = 53.7$ atm (with $dP_5/dt \sim 0\%/ms$).

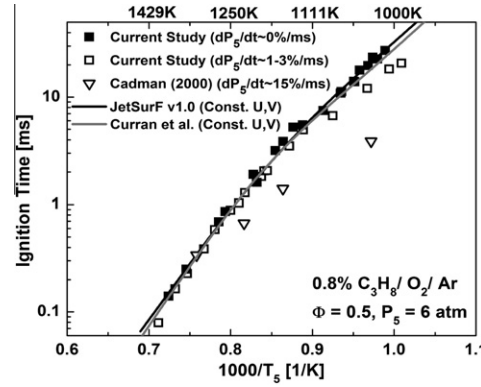

Fig. 5. Ignition delay times for 0.8% C_3H_8/ 8% O_2/ Ar mixture at $P_5 = 6$ atm, plotted at the initial post-shock T_5. Experimental data and calculated values from JetSurF v1.0 mechanism [10] and Curran et al. mechanism [5].

slightly colder initial temperature of 996 K is at least one-third shorter than the near-constant-volume measurement taken with the driver insert ($dP_5/dt \sim 0\%/ms$) at $T_5 = 1008$ K.

At temperatures less than 1100 K (or test times longer than 5 ms), a small pressure bump is observed in the LPST experiments (extending from about 5–8 ms in Fig. 3), even though a near-perfect tailoring condition has been achieved. This is primarily due to the fact that the contact surface is not actually a thin interface. Instead, it is a mixing zone, which has a finite width and is caused by non-ideal diaphragm bursting and boundary layer growth. Several strategies were attempted to eliminate such a pressure bump (see Davidson and Hanson [9]) with varying degrees of success, and eventually relatively small "bumps" were achieved typified by Fig. 3. In the current study, such pressure bumps do not affect the ignition delay time measurements significantly.

The low-pressure ignition delay times under both near-constant-volume conditions and conventional operation conditions are plotted at the initial reflected shock temperatures in Fig. 5, along with the calculated values from the JetSurF v1.0 mechanism of Sirjean et al. [10] and the Curran et al. mechanism of NUI Galway [5] under constant U, V constraints. The calculated values from both mechanisms are in excellent agreement at a pressure of 6 atm. At reflected shock temperatures less than 1250 K and pressures of 5.3–68 atm, an ignition time correlation for the current mixture can also be determined from a total of 50 near-constant-volume measurements with a R^2 value of 0.990: $\tau = 2.49 \times 10^{-5}\, P^{-0.89} \exp(15674/T)$, where the ignition time is in milliseconds, the pressure is in atmospheres, and the temperature is in kelvins. All data were then scaled by $P^{-0.89}$ to their nominal pressures. Note that both sets of measurements in the current study agree with each other at temperatures above 1110 K (i.e. at short test times). The discrepancy between the two measurement sets increases with decreasing temperature, but the constant U, V predictions remain in good agreement with the near-constant-volume ignition delay time measurements ($dP_5/dt = 0\%/ms$) at all temperatures. In contrast, the current data under conventional operation conditions ($dP_5/dt = 1$–$3\%/ms$) are shorter than the results from both models, starting at $T_5 = 1110$ K. For instance, the uncompensated conventional data is approximately 15 ms faster than the constant U, V models at $T_5 = 990$ K. Figure 5 also includes the previous ignition data from Cadman et al. [1] using the same test mixture. In the study performed by Cadman et al., the data were obtained at a reflected shock pressure of 5 atm. To perform a quantitative comparison, those ignition data were also scaled by $P^{-0.89}$ from $P_5 = 5$ to 6 atm. Notice that those previous measurements are substantially shorter than the current measurements and the constant U, V model predictions by at least a factor of 3. As mentioned previously, the inner diameter of the shock tube can have a significant impact on post-shock pressure variations. The Cadman et al. shock tube facility has an inner diameter of about 6 cm, which is much smaller than that of the Stanford low-pressure shock tube (14.13 cm). Based on their published pressure profiles (Fig. 5 from their publication), a relatively large post-shock pressure variation of approximately 15%/ms or higher was identified. Such large dP_5/dt can contribute to the substantial differences between the previous and current measurements. For kinetic modeling of reflected shock experiments, it is now clear that it is critically important to account for the post-shock

pressure variations in different shock tube facilities and that pressure growth rates be documented to allow proper modeling of experiments. Incorrect reaction pathways and rate coefficients may be inferred if uncompensated ignition data are fit by detailed mechanisms.

As discussed by Pang et al. [6], constant U, V modeling of low-temperature shock tube ignition delay times with finite dP_5/dt gives an unsatisfactory and incorrect representation of the actual chemical kinetics. A gasdynamic solver developed in our laboratory, named CHEMSHOCK, can be used to reasonably account for the non-ideal growth of pressure and temperature in the reflected shock region, up to the time of ignition, at least for simple reaction systems without significant pre-ignition heat release. CHEMSHOCK performs a two-step simulation process over each small time step: (1) constant U, V calculations, followed by (2) an assumed isentropic adjustment in conditions to recover the measured value of pressure at this time. This code has been validated against simulations from a one-dimensional reacting computational fluid dynamics code for a heptane/O_2/Ar mixture using a reduced mechanism, and against experimental data (such as gas temperature and water vapor concentration) in a hydrogen/O_2/Ar mixture by Li et al. [11]. The calculated ignition delay times from CHEMSHOCK using the JetSurF v1.0 mechanism and different pressure constraints ($dP_5/dt = 1.5\%/\text{ms}$ and 15%/ms) are displayed in Fig. 6. Here, the computed values are much closer to the uncompensated ignition delay times. By setting the post-shock pressure variation to 1.5%/ms, the model agrees with the conventional data in the current study within ±10% at lower temperatures. Additionally, the CHEMSHOCK modeling is able to capture the early roll-off behavior displayed by the previous ignition data from Cadman et al. (within 20%) at low temperatures after incorporating the published post-shock pressure increase of 15%/ms.

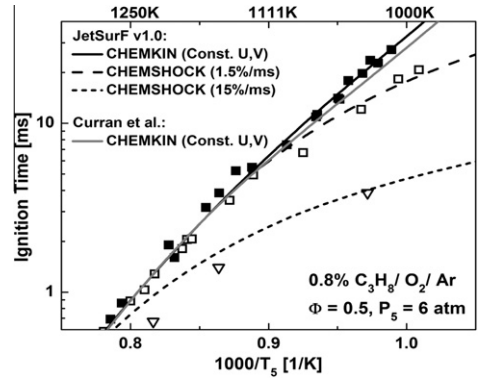

Fig. 6. Low-pressure experimental data and CHEMSHOCK modeling using JetSurF v1.0 mechanism.

Figure 7 shows the Arrhenius plots of lean propane ignition delay time measurements at elevated pressures of 24 and 60 atm, respectively. All experimental data were scaled to either 24 or 60 atm using $P^{-0.89}$. The measured facility-related rate-of-pressure change in the high-pressure shock tube was typically 6–7% per millisecond, which is much larger than observed in the low-pressure shock tube. Good agreement is found between the ignition delay data with dP_5/dt of ~0%/ms and ~6–7%/ms, at temperatures higher than 1110 K or when the ignition delay times are shorter than 1 ms. Notice, however, that the discrepancy in ignition delay data between conventional operation and near-constant-volume conditions is larger in the 24 atm experiments in the high-pressure shock tube than in the 60 atm experiments. For instance, at the initial reflected shock condition of $T_5 = 990$ K and $P_5 = 24$ atm, the uncompensated data are 50% shorter than

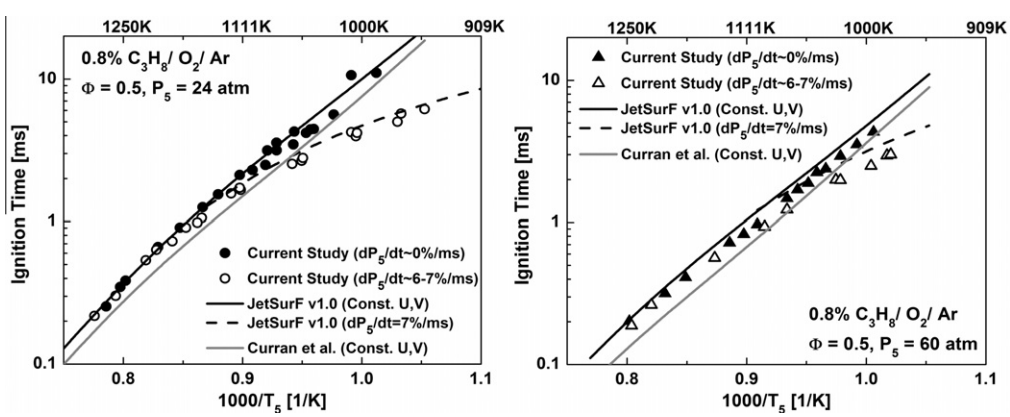

Fig. 7. High-pressure experimental data (at $P_5 = 24$ and 60 atm), along with CHEMKIN and CHEMSHOCK modeling using JetSurF v1.0 and Curran et al. mechanisms.

the near-constant-volume data. This is because the longer ignition delay times for the 24 atm data are more sensitive to changes in the temperature gradient than the shorter-ignition-time 60 atm data.

The compensated high-pressure experimental data in Fig. 7 are compared to predictions using both JetSurF and Curran et al. mechanisms under constant U, V constraints. At a nominal pressure of 24 atm, the computed ignition delay times from the JetSurF mechanism show very good agreement with the near-constant-volume measurements. At a nominal pressure of 60 atm, the constant U, V model predictions from the JetSurF mechanism agree reasonably well with the compensated data, but are consistently 15% longer than the measurements. However, the predicted values from the Curran et al. mechanism are at least 30% shorter than the near-constant-volume measurements at both pressures. Referring now to the uncompensated (conventional operation) results, all the experimental data are much shorter than the constant U, V model predictions at lower temperatures. The uncompensated data are at least three times shorter than the constant U, V model calculations (JetSurF) at $T_5 = 950$ K and $P_5 = 24$ atm, while they are 66% shorter at $T_5 = 996$ K and $P_5 = 60$ atm. Hence, the CHEMSHOCK model with the JetSurF mechanism was utilized to account for the non-ideal pressure rise in the reflected shock region. After incorporating a proper pressure constraint ($dP_5/dt = 7\%/\text{ms}$), the uncompensated data are successfully simulated at both pressures, as illustrated in Fig. 7. However, there does appear to be a small residual systematic difference between the measured and simulated ignition delay times in both the compensated and uncompensated 60 atm data. This may indicate a deficiency in the pressure-dependent chemistry in the JetSurF mechanism. A preliminary sensitivity analysis indicates that a 50% increase in the pressure-dependent hydrogen peroxide decomposition rate at 60 atm would eliminate this systematic difference.

4. Concluding remarks

Propane ignition delay times in a lean mixture (0.8% C_3H_8/8% O_2/Ar) were measured over the temperature range of 980–1400 K at nominal pressures of 6, 24 and 60 atm, under conventional shock tube operating conditions and under near-constant-volume conditions, using both the low- and high-pressure shock tubes. Under conventional operating conditions, a facility-related rate-of-pressure rise ($dP_5/dt \sim 1$–$7\%/\text{ms}$) is observed in the reflected shock region, introducing strong variations in the shock tube ignition delay time measurements. Under near-constant-volume test conditions, a flat pressure profile with small local variations was obtained using a driver-insert method and tailored driver gas mixtures. Good agreement in both types of measurements is found only at temperatures higher than 1110 K. As temperature is decreased, the discrepancy between the two types of measurements becomes larger at all pressures.

The near-constant-volume measurements of ignition delay times were found to be in good agreement with the predictions of the JetSurF v1.0 mechanism, using commonly employed constant U, V constraints. The ignition delay time measurements using conventional shock tube operation (with $dP_5/dt \sim 1$–$7\%/\text{ms}$) were also found to be in good agreement with detailed modeling, when CHEMSHOCK is used to include the effects of pressure and temperature variations. The new near-constant-volume data provide a valuable database for low-temperature propane mechanism development that is independent of fluid flow and heat transfer effects.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.131.

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